Hopkins Delivers: An Autonomous Drone Delivery Service

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Abstract

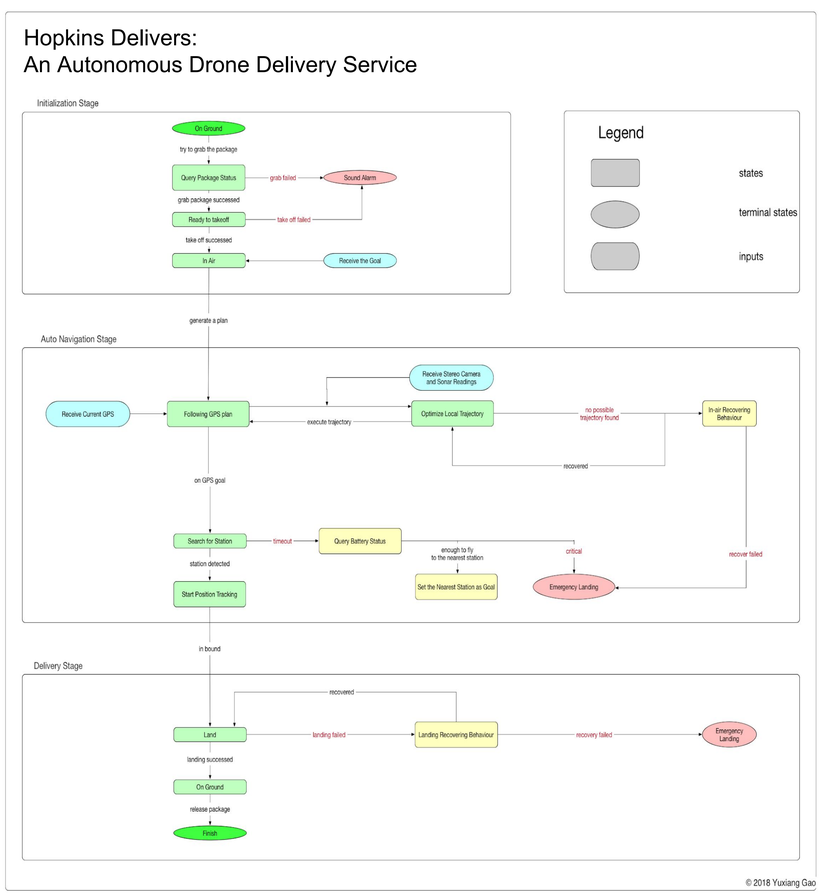
Research on the viability of commercial drone transportation and delivery has rapidly accelerated in the last five years. We at Hopkins Delivers have developed a prototype for an autonomous drone food delivery system on the Hopkins Homewood campus. Combining obstacle avoidance algorithms, proprietary safety features, and pick-up mechanisms, our system is built for ease-of-use for delivery vendors and students alike.

Concept Development

We surveyed Johns Hopkins students and affiliates to assess the food delivery habits of the Homewood campus and surrounding community. We then reached out to potential campus partners and identified landing sites. We also conducted industry research to determine key players and market positioning.

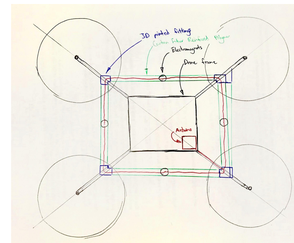
We chose to work with a commercially-available, developer-ready quadcopter, the DJI Matrice 100, as our drone body. After conducting trade studies and weight and power analyses, we moved forward with our initial prototype in three sub-teams: navigation, pick-up and delivery, and safety. The software is written in Robotics Operating System (ROS).

System Flow

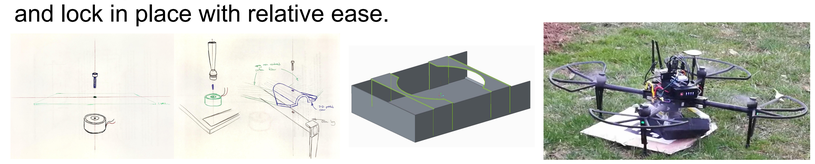


Pick-up and Delivery

Our prototype utilizes electromagnets to pick up the basket and a mechanical locking mechanism that secures the basket in-flight. We have several versions of the delivery basket intended for take out containers like pizza boxes and carry-out boxes.

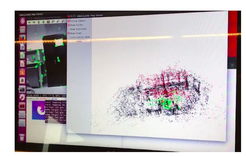


The basket is comprised of a carbon fiber reinforced polymer (CFRP) body with steel bars on top that helps the basket align and attach to the electromagnets, which in turn allows the mechanical mechanism to turn and lock in place with relative ease.



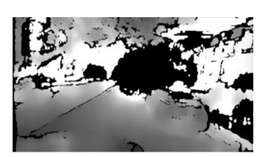
Navigation

An onboard ultrasound sonar system detects when an obstacle appears in the drone airline. The drone avoids obstacles based on data from two sonars which are driven by an Arduino and placed on both side of the drone.

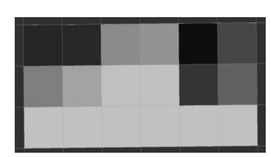


The Kalman Filter, an optimal estimation algorithm, is used to predict and correct the positiable location of obstacles.

Real time image



Depth image



Depth Histogram

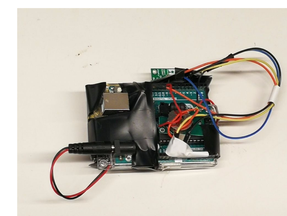
The drone avoids obstacles using feedback from the visual field. Based on depth images from a ZED camera, a specified distance threshold was set at 2000 mm. The software then divides the view into 12 visual receptive fields, calculating the sum of pixels, which are under 2000 mm in each visual field.

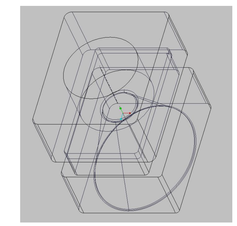
https://april.eecs.umich.edu/software/apriltag/

For precise landing, we programmed a monocular camera to recognize and track apriltags.

Safety

In the event of a drone malfunction while in flight, we programmed a siren system to warn audibly when the drone is in freefall and about to crash. We designed a controlling circuit using an Arduino. When the 6-axis accelerometer in the circuit detects certain acceleration on the z axis (detecting falling), the Arduino will trigger the 130 dB buzzer.

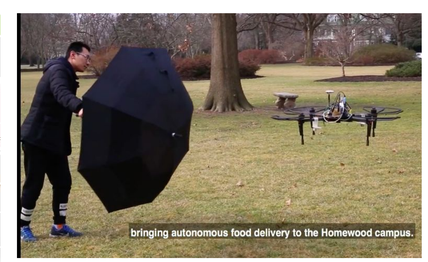


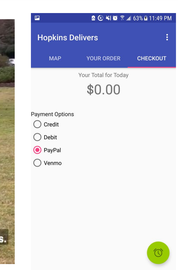


In order to mitigate damage to drone, package, and environment, we designed a parachute funnel system to slow the landing of the drone in freefall. A funnel tucks the parachute over a small opening. The bottom of the funnel is wider, so it can grab more air. Once the drone begins to fall, air rushes into the funnel. The smaller opening creates a force that pushes out and deploys the parachute.

We are also researching airbags as a safety mechanism. Once the drone drops, an accelerometer, running on the same code as the siren, could detect the falling motion and trigger an air cylinder to inflate the airbag. For testing, we are using standard airplane life vests that use 33g air cylinders on a small Parrot AR Drone, as opposed to the DJI Matrice 100.

Next Steps





Hopkins Delivers will continue to debug software, refine safety mechanisms, and improve weight and power budget. Towards marketing a commercially viable model, we also intend to deploy a mobile application for restaurant partners and customers alike, and form campus partnerships to ensure the successful launch of our product. We will also conduct user experience testing in order to better serve our customers.

Acknowledgments

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